Memory map

All addresses below are hexadecimal byte addresses. Loads sample sensors in MEM. Only a valid MEM-stage store can update RAM or an output latch. Reset clears actuator and timer state,& RAM contents are unspecified.

Address	Register	Access	Reset	Bits and behavior
00–7F	RAM	R/W	Unspecified	128 bytes
80–EF	Reserved	R zero / W ignored	0 read	No RAM or peripheral effect
F0	Packed inputs	Read only	External	bit 0 START; 2:1 MODE; 3 DOOR; 7:4 zero
F1	Water level	Read only	External	Unsigned bits 7:0
F2	Temperature	Read only	External	Bits 7:0
F3	Drum RPM	Read only	External	Bits 7:0
F4	Inlet valve	Write only	0	bit 0 request
F5	Heater	Write only	0	bit 0 request
F6	CPU motor enable	Write only	0	bit 0 request; separate safety gating
F7	Motor direction	Write only	0	bit 0 direction
F8	Drain pump	Write only	0	bit 0 request
F9	Door lock	Write only	0	bit 0 request; does not close door sensor
FA	Display	Write only	00	Bits 7:0 status
FB	Buzzer	Write only	0	bit 0 completion
FC	Timer COUNT	Read only	00	8-bit count, wraps FF to 00
FD	Timer LIMIT	R/W	00	Nonzero enables match comparison
FE	Timer CTRL	R/W	00	bit 0 enable; write bit 1 clear pulse
FF	Timer STATUS	Read only	00	bit 0 sticky MATCH; 7:1 zero

Reads of F4–FB return zero, observe their dedicated output pins/probes. Upper bits of one-bit actuator writes are ignored. Writes to sensors, COUNT and STATUS are ignored. All 16 peripheral addresses are assigned; any unimplemented peripheral selection returns zero and ignores writes.

CTRL reads only the stored enable bit-its clear bit reads as zero. Write 03 to clear COUNT/MATCH and enable, write 02 to clear and disable & write 01 to enable without clear & write 00 to disable. Clear wins over increment/comparison at that edge. MATCH is sticky until clear/reset. Writing LIMIT alone never sets MATCH retroactively.